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The scheduling of adolescence with Netrin-1 and UNC5C

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Daniel Hoops, Robert F. Kyne, Samer Salameh, Elise Ewing, Alina T. He, Taylor Orsini, Anais Durand, Christina Popescu, Janet M. Zhao, Kelcie C. Schatz, LiPing Li, Quinn E. Carroll, Guofa Liu, Matthew J. Paul, Cecilia Flores

Department of Psychiatry, McGill University, Montréal, Quebec, Canada • Douglas Mental Health University Institute, Montréal, Quebec, Canada • Neuroscience Program, University at Buffalo, SUNY, New York, USA • Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada • Department of Psychology, University at Buffalo, SUNY, New York, USA • Department of Biological Sciences, University of Toledo, Ohio, USA • Department of Neurology and Neurosurgery, McGill University, Montréal, Quebec, Canada

Abstract

Dopamine axons are the only axons known to grow during adolescence. Here, using rodent models, we examined how two proteins, Netrin-1 and its receptor, UNC5C, guide dopamine axons towards the prefrontal cortex and shape behaviour. We demonstrate in mice (*Mus musculus*) that dopamine axons reach the cortex through a transient gradient of Netrin-1 expressing cells – disrupting this gradient reroutes axons away from their target. Using a seasonal model (Siberian hamsters; *Phodopus sungorus*) we find that mesocortical dopamine development can be regulated by a natural environmental cue (daylength) in a sexually dimorphic manner – delayed in males, but advanced in females. The timings of dopamine axon growth and UNC5C expression are always phase-locked. Adolescence is an ill-defined, transitional period; we pinpoint neurodevelopmental markers underlying this period.

eLife assessment

This study addresses an **important** question using approaches that link molecular, circuit, and behavioral changes. The findings that Netrin-1 and UNC5c can guide dopaminergic innervation from the nucleus accumbens to the cortex during adolescence are **solid**. The data showing that the onset of Unc5 expression is sexually dimorphic in mice, and that in Siberian hamsters, environmental effects on development are also sexually dimorphic are **solid**. While this work is novel and on an interesting, understudied topic, the reviewers also identified significant gaps in experimental data and agree that support for the main claims is **incomplete**.

Introduction

Adolescence is a critical developmental period involving dramatic changes in behaviour and brain anatomy. The prefrontal cortex, the brain region responsible for our most complex cognitive functions, is still establishing connections during this time (Gogtay et al., 2004; Petanjek et al., 2011; Sowell et al., 2004). The trajectory of prefrontal cortex development in adolescence determines the vulnerability or resilience of individuals to adolescent-onset psychiatric diseases (Fuhrmann et al., 2015; Keshavan et al., 2014; Kessler et al., 2007; 2005; Lee et al., 2014). The age at which this adolescent development occurs therefore represents a critical window during which the brain is particularly susceptible to environmental influences. Traditionally, the onset of adolescence is thought to coincide with puberty (Hollenstein and Lougheed, 2013). In humans, the age of pubertal onset has been advancing throughout the 19th, 20th and 21st centuries, and environmental influences, such as nutrition, can pathologically alter the age of puberty (Wolf and Long, 2016). However, it remains entirely unknown whether the neural and cognitive maturational processes of adolescence can also be plastic. Here, for the first time, we examine how the timing of adolescence is programmed in the brain, and whether this timing can be plastic in response to a natural environmental cue, in parallel with pubertal plasticity.

Dopamine innervation to the prefrontal cortex increases substantially across adolescence and psychopathologies of adolescent origin prominently feature dopamine dysfunction. Evidence continues to emerge that protracted dopamine innervation is a key neural process underlying the cognitive and behavioural changes that characterize adolescence (Larsen and Luna, 2018). The mesocorticolimbic dopamine system – which includes the prefrontal cortex – is unique because not only are connections being formed and lost during adolescence, but there is also long-distance displacement of dopamine axons between brain regions. At the onset of adolescence, both mesolimbic and mesocortical dopamine axons innervate the nucleus accumbens in rodents, but the mesocortical axons leave the accumbens and grow towards the prefrontal cortex during adolescence and early adulthood (Hoops et al., 2018; Reynolds et al., 2018a). This is the only known case of axons growing from one brain region to another so late during development (Hoops and Flores, 2017).

The prolonged growth trajectory renders mesocortical dopamine axons particularly vulnerable to disruption. Environmental insults during adolescence (e.g. drug abuse) alter the extent and organization of dopamine innervation in the prefrontal cortex, leading to behavioural and cognitive changes in mice throughout adulthood (Drzewiecki and Juraska, 2020; Hoops and Flores, 2017; Reynolds and Flores, 2021). These changes often involve cognitive control, a prefrontal function that develops in parallel with dopamine innervation to the cortex in adolescence (Luna et al., 2015). Disruption of dopamine innervation frequently seems to result in “immature” cognitive control persisting through adulthood (Larsen and Luna, 2018).

Here, we examine the guidance of growing dopamine axons to the prefrontal cortex, and its timing. The guidance cue molecule Netrin-1, upon interacting with its receptor DCC, determines *which* dopamine axons establish connections in the nucleus accumbens and which ones leave this region to grow to the prefrontal cortex (Hoops and Flores, 2017; Reynolds and Flores, 2021). We hypothesized that the answers to *how* and *when* this extraordinary developmental feat is achieved may also lie in the Netrin-1 signalling system.

Part 1: Netrin-1 "paves the way" for dopamine axons in adolescence

To identify the route by which dopamine axons grow from the nucleus accumbens to the medial prefrontal cortex we visualized dopamine axons in the adult mouse forebrain. We observed that dopamine axons medial to the nucleus accumbens form a distinct bundle oriented dorsally towards the cortex (Figure 1A,B). Individual fibres can be seen crossing the boundary between the nucleus accumbens shell and this bundle (Figure 1C). We hypothesized that these are the fibres that grow to the prefrontal cortex during adolescence. If this is correct, the number of dopamine axons in the bundle should continue to increase until adulthood. To test this, we used a modified unbiased stereological approach (Kim et al., 2011) where axons are counted only if they crossed the upper and lower bounds of a counting probe. We also measured the average width of the dopamine axon bundle. We found, in both male and female mice, that the density of dopamine axons does not change between adolescence (21 days old) and adulthood (75 days old; Figure 1D). However, the width of the dopamine axon bundle does change, increasing between adolescence and adulthood (Figure 1E). These results indicate that the total number of dopamine axons passing through this bundle increases over adolescence and that dopamine axons grow to the medial prefrontal cortex via this route.

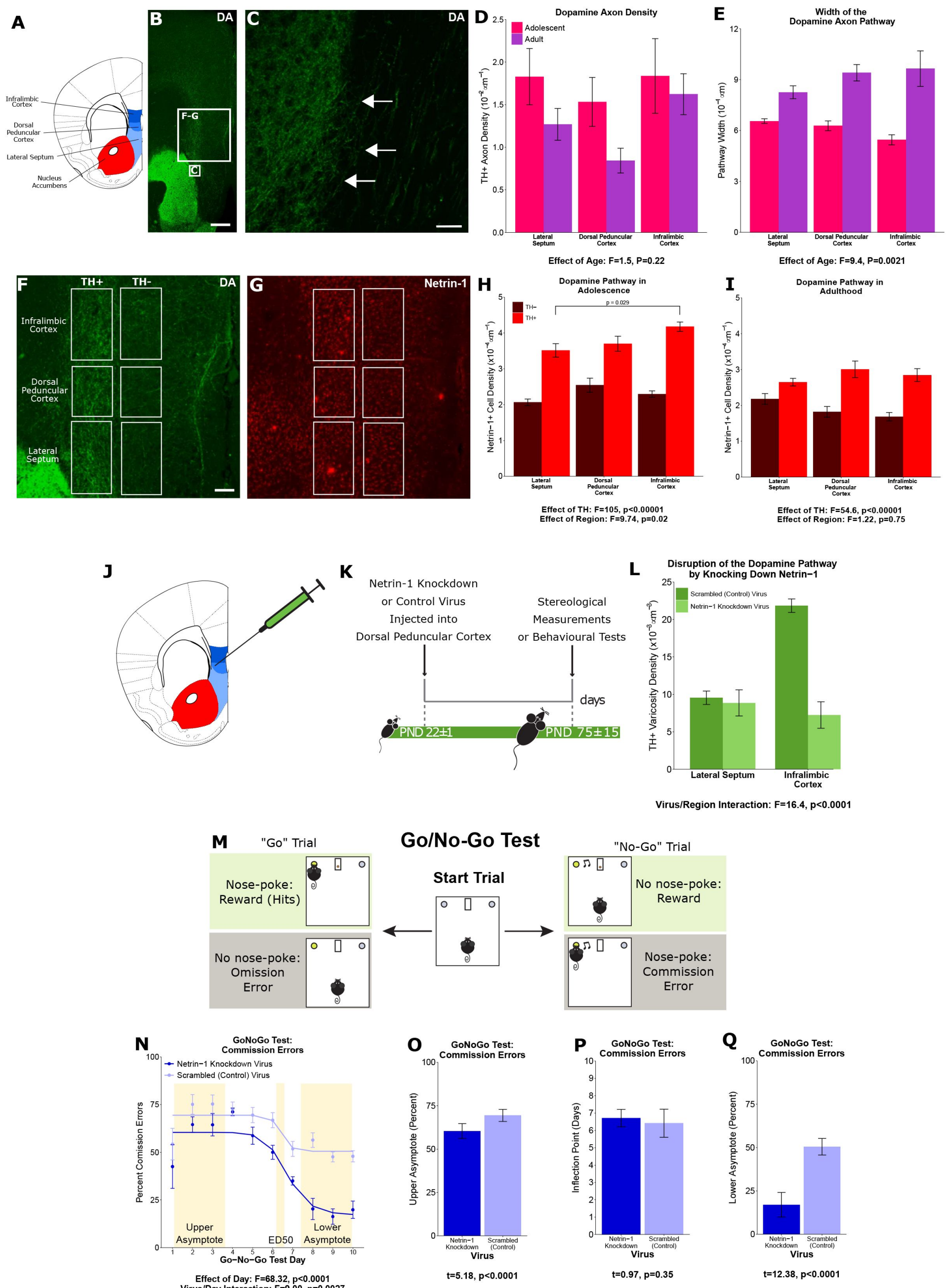

Figure 1.

A "pathway" of Netrin-1 expressing cells "paves the way" for dopamine (DA) axons growing from the nucleus accumbens to the medial prefrontal cortex during adolescence. **A**, The brain regions containing the bundle of dopamine fibres passing to the medial prefrontal cortex are highlighted in a line drawing of a coronal mouse brain section derived from Paxinos and Franklin (Paxinos and Franklin, 2013). **B**, An image of a coronal section through the forebrain of an adult mouse at low magnification (4x). Green fluorescence indicates immunostaining for tyrosine hydroxylase (TH), a marker for dopamine. The smaller and larger white squares indicate the regions enlarged in panel C and panels F & G, respectively. Scale bar = 500 μm. **C**, The nucleus accumbens (left) is densely packed with dopamine axons (in green). Some of these axons can be observed extending from the nucleus accumbens medially into a bundle of dopamine fibres (white arrows). Scale bar = 10 μm. **D**, Modified stereological quantification revealed no significant difference in dopamine axon density in the fibre bundle between adolescence (21 days old) and adulthood (75 days old). Mixed-effects ANOVA, effect of age: F=1.53, p=0.22; region by age interaction: F=1.44, p=0.49. **E**, The average width of the fibre bundle increased significantly from adolescence to adulthood, revealing that there is an increase in the total number of fibres passing to the medial prefrontal cortex during this period. Mixed-effects ANOVA, effect of age: F=9.45, p=0.0021; region by age interaction: F=5.74, p=0.057. **F**, In order to quantify the Netrin-1 positive cells along the dopamine fibre bundle, the bundle was contoured in each region, and a contour of equal area was placed medial to the fibre bundle as a negative control. Scale bar = 200 μm. **G**, Using quantitative stereology, Netrin-1 positive cell density was determined along and adjacent to the fibre bundle for each region. Red fluorescence indicates immunostaining for Netrin-1.**H**, In adolescent mice there are more Netrin-1 positive cells along the dopamine fibre bundle ("TH+") than medial to it ("TH-"). This is what we refer to as the "Netrin-1 pathway". Along the pathway, there is a significant increase in Netrin-1 positive cell density in regions closer to the medial prefrontal cortex, the innervation target. Mixed-effects ANOVA, effect of TH: F=105, p<0.0001. Effect of region: F=9.74, p=0.021. A post-hoc Tukey Test revealed a difference (p = 0.029) between the densities of the lateral septum and infralimbic cortex, but only within the dopamine fibre bundle. **I**, In adult mice the Netrin-1 pathway is maintained, however there is no longer an increasing density of Netrin-1 positive cells towards the medial prefrontal cortex. Mixed-effects ANOVA, effect of TH: F=54.56, p<0.0001. Effect of region: F=1.22, p=0.75. **J,** The virus injection location within the mouse brain. A Netrin-1 knockdown virus, or a control virus, was injected into the dopamine fibre bundle at the level of the dorsal peduncular cortex. **K,** Our experimental timeline: at the onset of adolescence a Netrin-1 knockdown virus, or a control virus, was injected, and then in adulthood the mice were sacrificed and stereological measurements taken. **L,** Dopamine varicosity density was quantified in the region below the injection site, in the lateral septum, and in the region above the injection site, in the infralimbic cortex. There was a significant decrease in dopamine varicosity density only in the infralimbic cortex. Mixed-effects ANOVA, virus by region interaction: F=16.41, p<0.0001. **M,** The experimental set-up of the final (test) stage of the Go/No-Go experimental protocol. A mouse that has previously learned to nose-poke for a reward in response to a visual cue (illuminated nose-poke hole), must now inhibit this behaviour when the visual cue is paired with an auditory cue (acoustic tone). **N,** Mice injected with the Netrin-1 knockdown virus show improved action impulsivity than controls because they incur significantly fewer commission errors across the Go/No-go task. Mixed-effects ANOVA, effect of day: F=68.32, p<0.0001. Day by virus interaction: F=9.00, p=0.0027. A sigmoidal curve is fit to each group of mice to determine how the two groups differ. Points indicate group means and error bars show standard error means. **O**, During the first days of Go/No-Go testing, both groups incur commission errors with high frequency, but, the Netrin-1 knockdown group has fewer errors than the control group (t-test, t=5.18, p<0.0001. **P,** The ED50 – the inflection point in each sigmoidal curve – are not different between groups, indicating that all mice improve their ability to inhibit their behavior at around the same time (T-test, t=0.97, p=0.35). **Q,** Mice microinfused with the Netrin-1 knockdown virus incur substantially fewer commission errors in the last days of the Go/No-Go task compared to mice injected with the control virus (T-test, t=12.38, p<0.0001). For all barplots, bars indicate group means and error bars show standard error means.

Next, we focussed on Netrin-1, a secreted protein that acts as a guidance cue to growing axons and is important for dopamine axon targeting in the nucleus accumbens in adolescence (Cuesta et al., 2020). Using unbiased stereology, we quantified the number of Netrin-1 expressing cell bodies along the dopamine axon route, and in an adjacent medial region as a control (Figure 1 F, G). We found that in adolescence there are more Netrin-1 positive neurons within the dopamine axon route than adjacent to it. Furthermore, along the axon route the density of Netrin-1 positive cells increases towards the medial prefrontal cortex, forming a dorsoventral gradient (Figure 1H). In adulthood, there remains a higher density of Netrin-1 positive cells along the dopamine route compared to the adjacent region, however the dorsoventral gradient is no longer present (Figure 1I).

To determine if Netrin-1 along the dopamine axon route is necessary for axon navigation, we silenced Netrin-1 expression in the dorsal peduncular cortex, the transition region between the septum and the medial prefrontal cortex, at the onset of adolescence (Figure 1J,K). In adulthood we quantified the number of dopamine axon terminals in the regions below and above the injection site. Silencing Netrin-1 did not alter dopamine terminal density below the injection, in the lateral septum (Figure 1L). In the infralimbic cortex, which is the first prefrontal cortical region the axons reach after the injection site, terminal density was reduced in the Netrin-1 knock-down group compared to controls (Figure 1L). The knock-down appears to erase the Netrin-1 path to the prefrontal cortex, resulting in dopamine axons losing their way and failing to reach their correct innervation target. We conclude that Netrin-1 expressing cells “pave the way” for the mesocortical dopamine axons.

We next examined how the Netrin-1 path may be important for behaviour. The dopamine input to the prefrontal cortex is a key factor in the transition from juvenile to adult behaviours that occurs in adolescence. We hypothesized that cognitive processes involving mesocortical dopamine function would be altered when these axons are misrouted in adolescence. To test our hypothesis we used the Go/No-Go behavioural paradigm. This test quantifies inhibitory control, which matures in parallel with the innervation of dopamine axons to the prefrontal cortex in adolescence (Casey et al., 2008; Klune et al., 2021; Luna et al., 2015; Paus, 2005; Reynolds and Flores, 2021; Spear, 2000) and it is impaired in adolescent-onset disorders like depression and schizophrenia (Catts et al., 2013; Clementz et al., 2016; McTeague et al., 2016; Millan et al., 2012).

At the onset of adolescence, we injected the Netrin-1 silencing, or a scrambled control virus, bilaterally into the dorsal peduncular cortex; in adulthood we tested the mice in the Go/No-Go task. This paradigm first involves discrimination learning and reaction time training (Supplementary Figure 1), followed by a Go/No-Go test consisting of "Go" trials where mice respond to a cue as previously trained and "No-Go" trials where mice must abstain from responding to the cue (Figure 1M). Correct responses to both trial types are reinforced with a food reward. We quantified the percent of "No-Go" trials where the mice incorrectly responded to the cue ("Commission Errors") and the percent of "Go" trials where the mice correctly responded ("Rewards" or "Hits"; Supplementary Figure 2). The ability of mice to respond correctly overall to both trial types is quantified as the Correct Response Rate (Supplementary Figure 3) (Cuesta et al., 2019; Reynolds et al., 2018a; 2018b; Vassilev et al., 2021).

Mice injected with the Netrin-1 silencing virus differed from controls in their performance during "No-Go" trials. As the mice learn to withhold their responses over the course of the test, the number of commission errors they made in No-Go trials decreased in a sigmoidal fashion (Figure 1N). The upper and lower asymptotes of the sigmoidal curve quantify the number of commission errors committed during early and late test days, respectively, while the inflection point (ED50) indicates when mice start improving their ability to inhibit their behavior. Already at the start of the Go/No part of the task, the Netrin-1 silencing group make slightly fewer commission errors (Figure 1O) than control groups, although both groups begin to improve in the No-Go task at around the same time. However, the Netrin-1 silencing group achieved a substantially higher level of behavioural inhibition, quantified as a lower percentage of commission errors in the last test days (Figure 1Q), indicating an improved ability to withhold their behaviour on cue. These behavioural results demonstrate that the maturation of action impulsivity is sensitive to the organization of the ventro-dorsal Netrin-1 path that guides mesocortical dopamine axon growth. Deviations in this route lead to striking changes in the cognitive development that is characteristic of adolescence.

Part 2: UNC5C expression coincides with the onset of adolescence

When axons leave the nucleus accumbens during adolescence, they follow a Netrin-1 "path" through intermediate brain regions to reach their intended innervation target. However, only a small subset of the dopamine axons that have reached the nucleus accumbens by early adolescence leave; the vast majority stay and form connections in the accumbens throughout life (Reynolds et al., 2018a). The "decision making" process of whether to "stay" (in the accumbens) or "go" (to the cortex via the Netrin-1 path) happens during a narrow developmental window at the onset of adolescence (Reynolds et al., 2018b). It remains unknown how the timing of this process is determined.

In adolescence, dopamine neurons begin to express the repulsive Netrin-1 receptor UNC5C, and reduction in UNC5C expression appears to cause growth of mesolimbic dopamine axons to the prefrontal cortex (Auger et al., 2013; Manitt et al., 2010; Pokinko et al., 2015). Using immunohistochemistry, we assessed the expression of UNC5C on nucleus accumbens dopamine axons across development. In male mice, we found little expression of UNC5C on dopamine axons at the onset of adolescence (Figure 2A), while we did find UNC5C expression on dopamine axons in adults (Figure 2B). Remarkably, when we assessed this in females, we found dopamine axons already expressing UNC5C in the nucleus accumbens at the onset of adolescence (Figure 2D), similar to adult females (figure 2E), indicating that the onset of UNC5C expression on dopamine axons in the nucleus accumbens is sexually dimorphic, with an earlier emergence in females. We examined the nucleus accumbens in pre-adolescent female mice and indeed found little UNC5C expression on dopamine axons (Figure 2C). The onset of UNC5C expression in mesocorticolimbic dopamine axons is therefore peri-adolescent but occurs earlier in females than in males, consistent with the earlier emergence of adolescence in female rodents and the earlier onset of adolescence and puberty in humans (Wolf and Long, 2016).

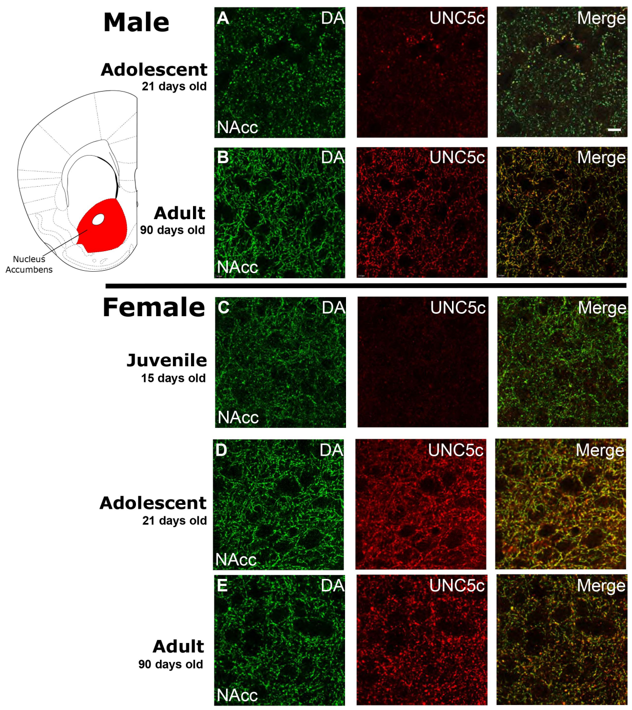

Figure 2.

The age of onset of UNC5C expression by dopamine axons in the nucleus accumbens of mice is sexually dimorphic. Images are representative of observed immunofluorescence patterns in the nucleus accumbens core (approx. location highlighted in the coronal mouse brain section Plate 19, modified from (Paxinos & Franklin, 2013)). For each row, six individuals were sampled. In males (**A-B**), UNC5C expression on dopamine fibres in the nucleus accumbens appears during adolescence. **A,** At the onset of adolescence (21 days old) dopamine fibres do not express UNC5C. Scale bar = 10 µm. **B,** By adulthood (90 days old), dopamine fibres express UNC5C. In females (**C-E**), UNC5C expression on dopamine fibres in the nucleus accumbens appears prior to adolescence. **C,** In juvenile (15 day old) mice, there is no UNC5C expression on dopamine fibres. **D,** By adolescence, dopamine fibres express UNC5C. **E,** In adulthood, dopamine fibres continue to express UNC5C.

Part 3: Environmental control of the timing of adolescence

We hypothesize that at the emergence of adolescence, UNC5C expression by dopamine axons in the nucleus accumbens signals the initiation of the growth of dopamine axons to the prefrontal cortex. We therefore examined whether the developmental timings of UNC5C expression and dopamine innervation of the prefrontal cortex are similarly affected by an environmental cue known to delay pubertal development in seasonal species.

Siberian hamsters (*Phodopus sungorus*) regulate many aspects of their behavior and physiology to meet the changing environmental demands of seasonality (Paul et al., 2008; Stevenson et al., 2017). In winter, they increase the thickness of their fur, exchange their brown summer coats for white winter ones, and undergo a daily torpor to conserve energy (Scherbarth and Steinlechner, 2010). In addition, adults suppress reproduction and juveniles delay puberty (Pevet, 1988; Yellon and Goldman, 1984), including developmental changes in gonadotropin releasing hormone neurons in the hypothalamus (Buchanan and Yellon, 1991; Heywood and Yellon, 1997). This seasonal plasticity is regulated by long or short periods of daylight (Heldmaier and Steinlechner, 1981; Hoffmann, 1978) and raises the possibility that aspects of adolescent development are sensitive to these environmental cues. To our knowledge, adaptive variation in the timing of adolescent neural development has never been recorded in any animal.

Here, we tested whether day length regulates *when* dopamine axons grow to the cortex, and whether the timing of UNC5C expression in the nucleus accumbens and adolescent changes in exploratory behavior are similarly affected.

3.i The seasonality of adolescence

Male hamsters were examined at three ages: 15 days old (±1), 80 days old (±10), and 215 days old (±20). We compared the density of the dopamine innervation to the medial prefrontal cortex in male hamsters housed under lighting conditions that replicate summer daylengths (long days, short nights) or winter daylengths (short days, long nights) (Figure 3A,B). We will refer to these two groups as "summer hamsters" and "winter hamsters" to emphasize the natural stimulus we are replicating in the laboratory environment. We confirmed that puberty is delayed in male winter hamsters compared to summer hamsters in the present experiment by measuring their gonadal weights across ages (Supplementary Figure 4).

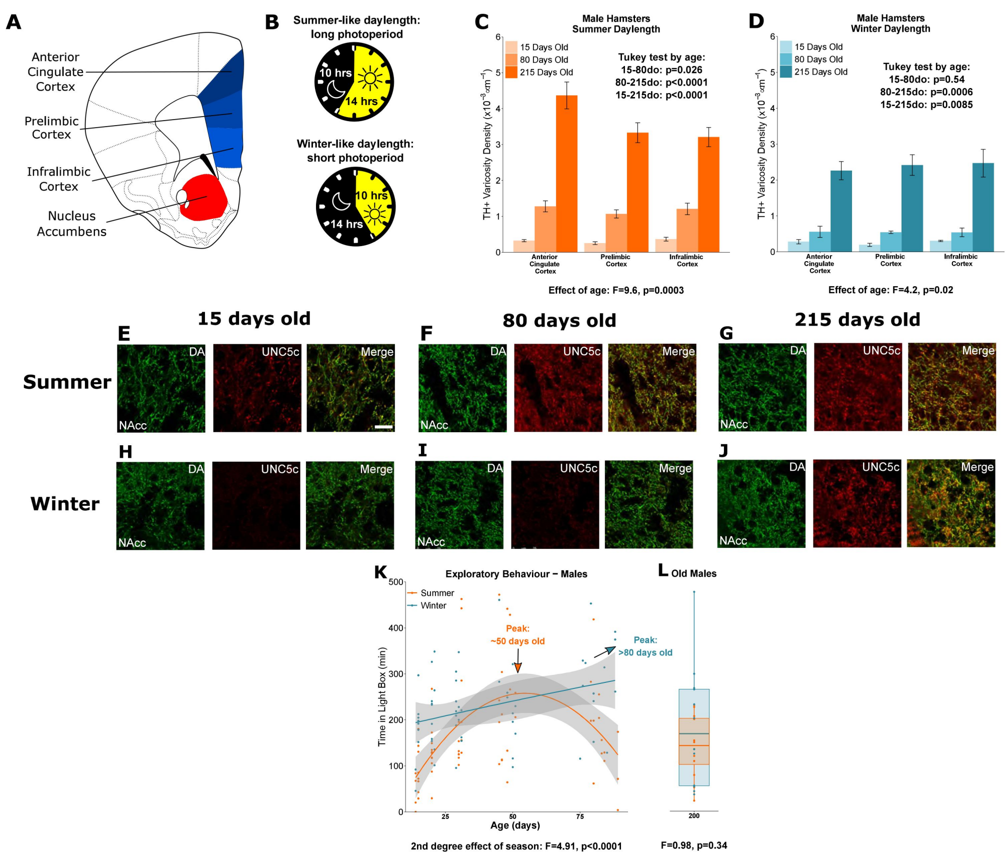

Figure 3.

Plasticity of adolescent development in male Siberian hamsters according to seasonal phenotype. All results illustrated in this figure refer to results in male hamsters. **A,** Dopamine innervation was quantified in three subregions of the medial prefrontal cortex, highlighted in blue. UNC5C expression was examined in the nucleus accumbens, highlighted in red. Line drawing of a coronal section of the mouse brain was derived from Plate 19 of Paxinos and Franlkin (Paxinos and Franklin, 2013). **B,** Hamsters were housed under either summer-mimicking long days and short nights ("summer hamsters") or winter-mimicking short days and long nights ("winter hamsters"). **C,** In male hamsters housed under a summer-mimicking daylength there is an increase in dopamine varicosity density in the medial prefrontal cortex between 15 and 80 days old. Mixed-effects ANOVA, effect of age: F=9.6, p=0.000255. Tukey Test, 15-80 days old (do): p=0.026; 80-215do: p<0.0001; 15-215do: p<0.0001. **D,** In male hamsters housed under a winter-mimicking daylength there is no increase in dopamine varicosity density until hamsters have reached 215 days of age. Mixed-effects ANOVA, effect of age: F=4.17, p=0.0205. Tukey Test, 15-80do: p=0.54; 80-215do: p=0.0006; 15-215do: p=0.0085. **E,** At 15 days old, dopamine axons in the nucleus accumbens of male summer-daylength hamsters largely do not express UNC5C. Scale bar = 20um (bottom right). **F-G,** At 80 (**F**) and 215 (**G**) days old, dopamine axons in the nucleus accumbens express UNC5C. **H-I,** At 15 (**H**) and 80 (**I**) days old, dopamine axons in the nucleus accumbens of male winter hamsters largely do not express UNC5C. **J,** By 215 days old there is UNC5C expression in dopamine axons in the nucleus accumbens of male winter hamsters. **E-J**, Representative images of the nucleus accumbens shell, 6 individuals were examined per group. **K,** Male hamsters house under a summer-mimicking daylength show an adolescent peak in exploratory behaviour in the light/dark box apparatus. Instead, those raised under a winter-mimicking photoperiod show a steady increase in exploratory behaviour over the same age range. Arrows indicate the ages at which exploratory behaviour peaks in summer (orange) and winter (blue) hamsters. Polynomial regression, effect of season: $F=3.551$, $p=0.00056$. **L,** In male hamsters, at 215 days of age, there is no difference in exploratory behaviour between hamsters raised under summer and winter photoperiods. T-test, effect of season: $t=0.975$, $p=0.341$. For all barplots, bars indicate group means and error bars show standard error means.

In male summer hamsters, dopamine input density to the prefrontal cortex increases during adolescence, after 15 days old and before 80 days old (Figure 3C), consistent with dopamine axon growth in mice (Manitt et al., 2013; 2011; Reynolds et al., 2018a). Prefrontal cortex dopamine innervation in summer hamsters continues to increase after 80 days old (Figure 3C).

In male winter hamsters, dopamine innervation to the prefrontal cortex is delayed until after 80 days, which coincides with their delayed pubertal onset (Figure 3D, Supplementary Figure 2B). This is the first demonstration of an environmental cue shaping adolescent brain development.

We then examined UNC5C expression by dopamine axons in the nucleus accumbens in male summer and winter hamsters across age classes. UNC5C expression was apparent only after the onset of adolescence in summer hamsters (Figure 3E,F,G), as observed in male mice. However, UNC5C expression was delayed in male winter hamsters – this group did not show UNC5C expression in dopamine axons in the nucleus accumbens until after 80 days old (Figure 3H,I,J). This aligns with the delayed timing of mesocortical dopamine axon growth and pubertal onset in male winter hamsters and demonstrates that the emergence of UNC5C is a marker of adolescent onset in male mice.

A behaviour characteristic of adolescence is increased willingness to enter a novel environment, called exploratory behaviour (Arrant et al., 2013; Lynn and Brown, 2009); to measure this we used the light/dark test (Bourin and Hascoët, 2003). Time spent in the light compartment is dopamine-dependent (Bahi and Dreyer, 2019; Gao and Cutler, 1993) and peaks in adolescence (Arrant et al., 2013). We assessed the developmental profile of exploratory behavior in the light/dark box test in summer and winter hamsters across adolescence. In male summer hamsters, the exploratory behavior increases across adolescence, peaks around 50 days, then subsequently declines (Figure 3K). However, the adolescent increase in exploratory behavior is protracted in winter hamsters: across the age range examined we observe a gradual, consistent increase in exploratory behaviour rather than a peak and decline.

We next assessed a cohort of 215-day old hamsters, for which both summer and winter male hamsters have undergone puberty and exhibit high levels of dopamine innervation of the prefrontal cortex (Figure 3C,D,G,J, Supplementary Figure 2B). In these hamsters, we find no difference in exploratory behaviour between the male summer and winter groups (Figure 3L), demonstrating that, after 80 days, exploratory behaviour begins to decline in male winter hamsters and that by 215 days it has declined to the same level as in summer hamsters. Male hamsters raised under summer-mimicking long days and winter-mimicking short days both ultimately make the transition to the adult behavioral phenotype.

3.ii An extraordinary case of decoupling puberty and adolescence

In parallel with males, we conducted equivalent experiments in female hamsters (Figure 4A,B). Under a summer-mimicking daylength, dopamine innervation to the medial prefrontal cortex increases between 15 and 80 days old, similar to male summer hamsters (Figure 4C). There is no further increase in innervation density after 80 days old, consistent with earlier adolescent development in females observed in other rodent species (Juraska and Willing, 2017; Kopec et al., 2018; Reynolds and Flores, 2021; Spear, 2000; Westbrook et al., 2018). We confirmed that puberty is delayed in female winter hamsters compared to summer hamsters by measuring their uterine weights across ages (Supplementary Figure 2A).

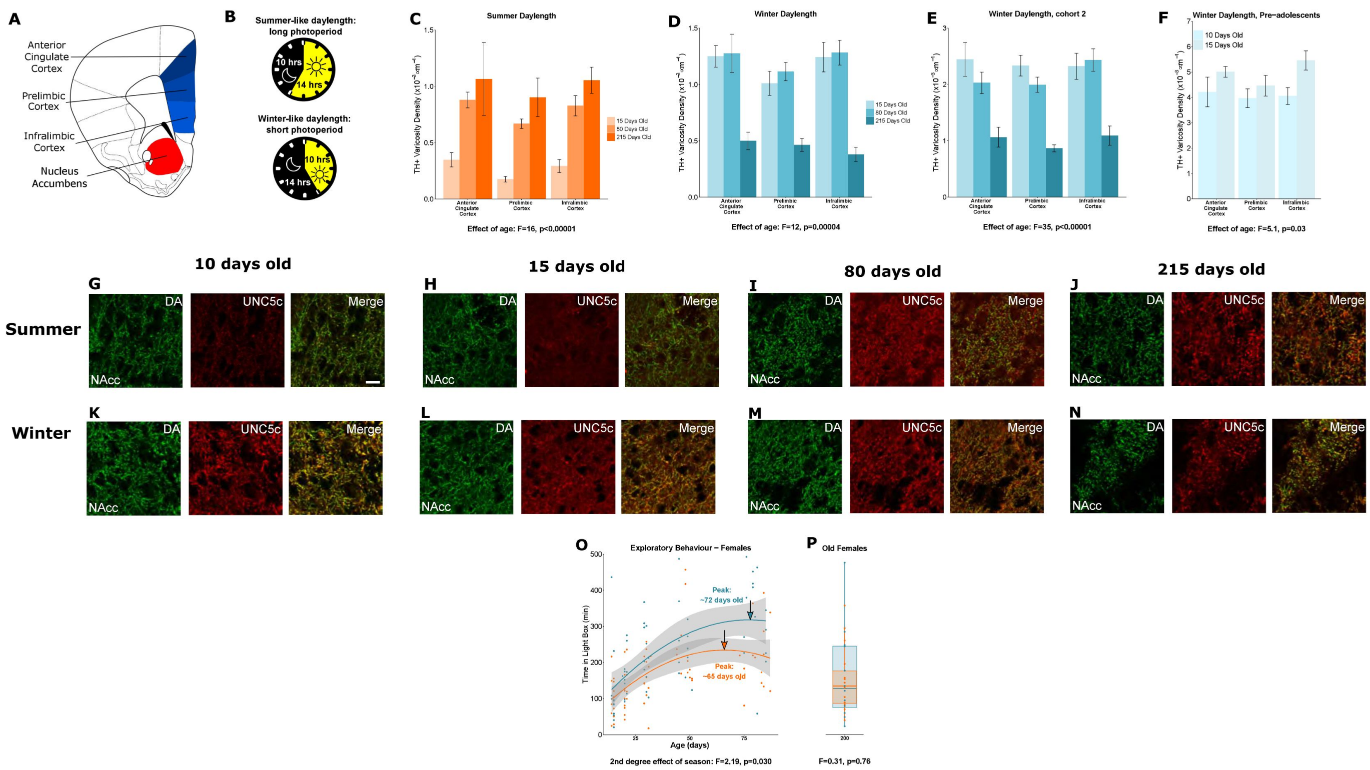

Figure 4

Plasticity of adolescent development in female Siberian hamsters according to seasonal phenotype. All results illustrated in this figure refer to results in female hamsters. **A,** Dopamine innervation was quantified in three subregions of the medial prefrontal cortex, highlighted here in blue. UNC5C expression was examined in the nucleus accumbens, highlighted in red. Line drawing of a coronal section of the mouse brain was derived from Paxinos and Franlkin (Paxinos and Franklin, 2013). **B,** Hamsters were housed under either a summer-mimicking or winter-mimicking daylength. **C,** In female hamsters housed under a summer daylength dopamine varicosity density in the medial prefrontal cortex increases between 15 and 80 days of age. Mixed-effects ANOVA, effect of age: $F=16.72$, $p<0.0001$ **D,** In female hamsters housed under a winter daylength there is no increase in dopamine varicosity density post-adolescence. Instead, there is a steep decline in density between 80 and 215 days of age. Mixed-effects ANOVA, effect of age: $F=12.33$, $p=0.000043$. **E,** As our results in panel D were unexpected, we replicated them with a second cohort of hamsters and found qualitatively identical results. Mixed-effects ANOVA, effect of age: $F=34.871$, $p<0.0001$. **F,** To try and determine when dopamine varicosities innervate the medial prefrontal cortex, we examined a cohort of 10- and 15-day-old hamsters. We found that varicosity density increases in the medial prefrontal cortex during this time, indicating that dopamine innervation to the medial prefrontal cortex is accelerated in female winter hamsters. Mixed-effects ANOVA, effect of age: $F=5.05$, $p=0.03$. **G-H,** In 10- and 15-day-old female summer hamsters there is little UNC5C expression in nucleus accumbens dopamine axons. Sample size: 4 (panel G) or 6 (panel H). **I-J,** By 80 days old (panel I), and continuing at 215 days old (panel J), dopamine axons in the nucleus accumbens express UNC5C in female summer hamsters. Sample sizes: 6. Scale bar = 20um (panel G bottom right). **K-N,** At all ages at which winter female hamsters were examined dopamine axons in the nucleus accumbens express

UNC5C in winter female hamsters. Sample sizes: 4 (panel K) or 6 (panels L-N). **O,** In female hamsters, those raised under summer and winter daylengths both show an increase in exploratory behaviour over time. The winter hamsters peak later compared to the summer daylength hamsters. Arrows indicate the ages at which exploratory behaviour peaks in summer (orange) and winter (blue) hamsters. Polynomial regression, effect of season: $F=3.305$, $p=0.00126$. **P,** In female hamsters, at 215 days of age, there is no difference in exploratory behaviour between hamsters raised under summer and winter photoperiods. T-test, effect of season: $t=0.309$, $p=0.76$. For all barplots, bars indicate group means and error bars show standard error means.

When housed under a winter-mimicking daylength, dopamine input density in the prefrontal cortex of female hamsters is *not* delayed as in males, but rather reaches adult levels prior to 15 days old (Figure 4D). We replicated this unexpected finding in a separate, independent cohort of female winter hamsters (Figure 4E). To our knowledge, this is the first time an intervention is shown to accelerate adolescent cortical development.

We then measured dopamine axon density in female winter hamsters at two earlier ages: 10 and 15 days old. Dopamine innervation increases during this period (figure 4F), well before normal adolescence and long before pubertal development. This is an extraordinary phenomenon: a key marker of adolescent neurodevelopment is accelerated and dissociated from puberty in female hamsters raised under winter-mimicking short days (Supplementary Figure 2A).

The early increase in prefrontal cortex dopamine terminals in winter females is followed by a dramatic reduction between 80 and 215 days old (Figure 4D,E). This overlaps with the delayed timing of puberty in these females (Butler et al., 2007; Huttenlocher, 1984; Koss et al., 2013; Petanjek et al., 2011). Under normal conditions, the effect of pruning on dopamine synapses is likely masked by the growth of new dopamine axons to the prefrontal cortex (Manitt et al., 2013; 2011; Reynolds et al., 2018a). In the case of female winter hamsters, we hypothesize that the growth of dopamine axons to the prefrontal cortex occurs early while synaptic pruning, including of dopamine synapses, appears to occur later. This leads to a remarkable dissociation between two cortical developmental processes that are normally simultaneous, the behavioural implications of which are unclear.

If the developmental onset of UNC5C expression determines the timing of dopamine innervation of the prefrontal cortex, then onset of UNC5C expression should also be advanced in female winter hamsters. Hence, we examined UNC5C expression at the same ages as we examined dopamine axon growth in female hamsters. At 10 and 15 days old, UNC5C expression is present *only* in the winter hamsters (Figure 4G,H,K,L), but at 80 and 215 days old, UNC5C expression is apparent in both summer and winter hamsters (Figure 4I,J,M,N).

We used the light/dark box test to examine potential exploratory behavioural implications of the extraordinary developmental trajectory we observed in the prefrontal cortex of female hamsters. In female summer and winter hamsters the adolescent increase and peak in exploratory behaviour occurs between the ages of 15 and 80 days, as it does in summer daylength males (Figure 4O). However, contrary to what we would expect, the peak in winter females is delayed compared to summer females. When we assessed an independent cohort of 215-day-old female hamsters, we found no difference in exploratory behaviour between groups (Figure 4R), indicating that, like males, female summer and winter hamsters both eventually reach the same adult level of exploratory behaviour.

In both sexes, hamsters housed under a summer-mimicking daylength showed an adolescent peak in exploratory behaviour at an age that we would predict based on results from other rodents (Arrant et al., 2013; Pietropaolo et al., 2004; Tanaka, 2015). When raised under a winter-mimicking daylength, hamsters of either sex show a protracted exploratory peak. In males, it is delayed beyond 80 days old, but the delay is substantially less in females. For dopamine and UNC5C neural markers, males delayed, whereas females advanced, adolescent neural development. Based on the delayed developmental profile in exploratory behavior, we predict that other adolescent neural changes are delayed in female winter hamsters. We hypothesize that the advanced dopamine/UNC5C neural development lessens the delay in peak exploratory behaviour of female winter hamsters compared to that of males.

Conclusion

Here we describe how the gradual growth of mesocortical dopamine axons marks adolescent development, and how this process uses guidance cues and is sensitive to sex and environment. Netrin-1 signalling provides the “stay-or-go” “decision making” conducted by dopamine axons that innervate the nucleus accumbens at the onset of adolescence (Cuesta et al., 2020). UNC5C expression by these dopamine axons marks the timing at which this decision is made. In mice, UNC5C expression coincides with sex differences in both adolescent and pubertal development. Females, which develop earlier, show earlier UNC5C expression in dopamine axons compared to males.

In hamsters, behavioural and developmental shifts in response to environmental cues occur in parallel with alterations in the timing of dopamine axon growth. As we show here, male hamsters raised under a winter-mimicking daylength delay not only puberty, but also adolescent dopamine and behavioural maturation. In contrast, female hamsters under identical conditions delay puberty but accelerate dopamine axon growth, a key marker of adolescent brain development. Behavioural shifts during adolescence appear to be delayed in these females, but less substantially than in male hamsters. Notably, under all conditions, the developmental timing of UNC5C expression corresponds to the timing of dopamine innervation of the prefrontal cortex.

In both mice and hamsters, the emergence of UNC5C expression coincides with the onset of dopamine axon growth to the prefrontal cortex, a key characteristic of the adolescent transition period. While previously we have shown that the Netrin-1 signalling in the nucleus accumbens is responsible for coordinating *whether* dopamine axons grow in adolescence (Reynolds and Flores, 2021), here we propose that Netrin-1 signalling is also key to determining *how* and *when* this marker of adolescence occurs.

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Author contributions

Conceptualization: DH, MJP, CF

Data Curation: DH

Formal Analysis: DH

Investigation: DH, RFK, SS, EE, AH, TO, AD, MG, CP, JZ, KCS, LPL, QEC

Methodology: DH, MJP, CF

Resources: GL, MJP, CF

Supervision: DH, MJP, CF

Validation: DH

Visualization: DH

Writing—original draft: DH

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Competing interests

Authors declare that they have no competing interests.

Data and code availability

All data and code use in these analyses are available through the Open Science Framework (DIO 10.17605/OSF.IO/DU3H4).

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Author information

Daniel Hoops
Department of Psychiatry, McGill University, Montréal, Quebec, Canada, Douglas Mental Health University Institute, Montréal, Quebec, Canada
ORCID iD: 0000-0002-5707-3513

Robert F. Kyne
Neuroscience Program, University at Buffalo, SUNY, New York, USA

Samer Salameh
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada

Elise Ewing
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada

Alina T. He
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada

Taylor Orsini
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada

Anais Durand
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada

Christina Popescu
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada
ORCID iD: 0000-0003-1738-4680

Janet M. Zhao
Douglas Mental Health University Institute, Montréal, Quebec, Canada, Integrated Program in Neuroscience, McGill University, Montréal, QC, Canada

Kelcie C. Schatz
Department of Psychology, University at Buffalo, SUNY, New York, USA

LiPing Li
Department of Psychology, University at Buffalo, SUNY, New York, USA

Quinn E. Carroll
Department of Psychology, University at Buffalo, SUNY, New York, USA

Guofa Liu
Department of Biological Sciences, University of Toledo, Ohio, USA

Matthew J. Paul
Neuroscience Program, University at Buffalo, SUNY, New York, USA, Department of Psychology, University at Buffalo, SUNY, New York, USA

Cecilia Flores
Department of Psychiatry, McGill University, Montréal, Quebec, Canada, Douglas Mental Health University Institute, Montréal, Quebec, Canada, Department of Neurology and Neurosurgery, McGill University, Montréal, Quebec, Canada

For correspondence: cecilia.flores@mcgill.ca

Editors

Reviewing Editor
Jun Ding
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Reviewer #1 (Public Review):

In this study, Hoops et al. showed that Netrin-1 and UNC5c can guide dopaminergic innervation from nucleus accumbens to cortex during adolescence in rodent models. They found that these dopamine axons project to the prefrontal cortex in a Netrin-1 dependent manner and knocking down Netrin-1 disrupted motor and learning behaviors in mice. Furthermroe, the authors used hamsters, a seasonal model that is affected by the length of daylight, to demonstrate that the guidance of dopamine axons is mediated by the environmental factor such as daytime length and in sex dependent manner. While this study highlighted the important roles of two neurodevelopmental markers, netrin-1 and UNC5C, in the projection of dopaminergic azons in the adolescence/adult brain, the major weakness is that the data are quite superficial and do not establish any definitive evidence to support the causal relationship between the expression of netrin-1 and UNC5C in the projection of dopaminergic axons remain unclear.
Below are several major concerns regarding the data presented in this manuscript:

1. Despite the well-established role of Netrin-1 and UNC5C axon guidance during embryonic commissural axons, it remains unclear which cell type(s) express Netrin-1 or UNC5C in the dopaminergic axons and their targets. For instance, the data in Figure 1F-G and Figure 2 are quite confusing. Does Netrin-1 or UNC5C express in all cell types or only dopamine-positive neurons in these two mouse models? It will also be important to provide quantitative assessments of UNC5C expression in dopaminergic axons at different ages.

2. Figure 1 used shRNA to knockdown Netrin-1 in the Septum and these mice were subjected to behavioral testing. These results, again, are not supported by any valid data that the knockdown approach actually worked in dopaminergic axons. It is also unclear whether knocking down Netrin-1 in the septum will re-route dopaminergic axons or lead to cell death in the dopaminergic neurons in the substantia nigra pars compacta?

3. Another issue with Figure1J. It is unclear whether the viruses were injected into a WT mouse model or into a Cre-mouse model driven by a promoter specifically expresses in dorsal peduncular cortex? The authors should provide evidence that Netrin-1 mRNA and proteins are indeed significantly reduced. The authors should address the anatomic results of the area of virus diffusion to confirm the virus specifically infected the cells in dorsal peduncular cortex.

4. The authors need to provide information regarding the efficiency and duration of knocking down. For instance, in Figure 1K, the mice were tested after 53 days post injection, can the virus activity in the brain last for such a long time?

5. In Figure 1N-Q, silencing Netrin-1 results in less DA axons targeting to infralimbic cortex, but why the Netrin-1 knocking down mice revealed the improved behavior?

6. What is the effect of knocking down UNC5C on dopamine axons guidance to the cortex?

7. In Figures 2-4, the authors only showed the amount of DA axons and UNC5C in NAcc. However, it remains unclear whether these experiments also impact the projections of dopaminergic axons to other brain regions, critical for the behavioral phenotypes. What about other brain regions such as prefrontal cortex? Do the projection of DA axons and UNC5c level in cortex have similar pattern to those in NAcc?

8. Can overexpression of UNC5c or Netrin-1 in male winter hamsters mimic the observations in summer hamsters? Or overexpression of UNC5c in female summer hamsters to mimic the winter hamster? This would be helpful to confirm the causal role of UNC5C in guiding DA axons during adolescence.

9. The entire study relied on using tyrosine hydroxylase (TH) as a marker for dopaminergic axons. However, the expression of TH (either by IHC or IF) can be influenced by other environmental factors, that could alter the expression of TH at the cellular level.

10. Are Netrin-1/UNC5C the only signal guiding dopamine axon during adolescence? Are there other neuronal circuits involved in this process?

11. Finally, despite the authors' claim that the dopaminergic axon project is sensitive to the duration of daylight in the hamster, they never provided definitive evidence to support this hypothesis.

Reviewer #2 (Public Review):

In this manuscript, Hoops et al., using two different model systems, identified key developmental changes in Netrin-1 and UNC5C signaling that correspond to behavioral changes and are sensitive to environmental factors that affect the timing of development. They found that Netrin-1 expression is highest in regions of the striatum and cortex where TH+ axons are travelling, and that knocking down Netrin-1 reduces TH+ varicosities in mPFC and reduces impulsive behaviors in a Go-No-Go test. Further, they show that the onset of Unc5 expression is sexually dimorphic in mice, and that in Siberian hamsters, environmental effects on development are also sexually dimorophic. This study addresses an important question using approaches that link molecular, circuit and behavioral changes. Understanding developmental trajectories of adolescence, and how they can be impacted by environmental factors, is an understudied area of neuroscience that is highly relevant to understanding the onset of mental health disorders. I appreciated the inclusion of replication cohorts within the study.

Reviewer #3 (Public Review):

This study from the Flores group aims at understanding neuronal circuit changes during adolescence which is an ill-defined, transitional period involving dramatic changes in behavior and anatomy. They focus on DA innervation of the prefrontal cortex, and their interaction with the guidance cue Netrin-1. They propose DA axons in the PFC increase in the postnatal period, and their density is reduced in a Netrin 1 knockdown, suggesting that Netrin abets the development of this mesocortical pathway. In such mice impulsivity gauged by a go-no go task is reduced. They then provide some evidence that Unc5c is developmentally regulated in DA axons. Finally they use an interesting hamster model, to study the effect of light hours on mesocortical innervation, and make some interesting observations about the timing of innervation and Unc5c expression, and the fact that females housed in winter day length conditions display an accelerated innervation of the prefrontal cortex. While this work is novel and on an interesting, understudied topic, several aspects need to be further consolidated, to make it more persuasive.

Main comments
1. Fig 1 A and B don't appear to be the same section level.
2. Fig 1C. It is not clear that these axons are crossing from the shell of the NAC.
3. Fig 1. Measuring width of the bundle is an odd way to measure DA axon numbers. First the width could be changing during adult for various reasons including change in brain size. Second, I wouldn't consider these axons in a traditional bundle. Third, could DA axon counts be provided, rather than these proxy measures.
4. TH in the cortex could also be of noradrenergic origin. This needs to be ruled out to score DA axons
5. Netrin staining should be provided with NeuN + DAPI; its not clear these are all cell bodies. An in situ of Netrin would help as well.
6. The Netrin knockdown needs validation. How strong was the knockdown etc?
7. If the conclusion that knocking down Netrin in cortex decreases DA innervation of the IL, how can that be reconciled with Netrin-Unc repulsion.
8. The behavioral phenotype in Fig 1 is interesting, but its not clear if its related to DA axons/signaling. IN general, no evidence in this paper is provided for the role of DA in the adolescent behaviors described.
9. Fig2 - boxes should be drawn on the NAc diagram to indicate sampled regions. Some quantification of Unc5c would be useful. Also, some validation of the Unc5c antibody would be nice.
10. "In adolescence, dopamine neurons begin to express the repulsive Netrin-1 receptor UNC5C, and reduction in UNC5C expression appears to cause growth of mesolimbic dopamine axons to the prefrontal cortex".....This is confusing. Figure 2 shows a developmental increase in UNc5c not a decrease. So when is the "reduction in Unc5c expression" occurring?

11. In Fig 3, a statistical comparison should be made between summer male and winter male, to justify the conclusions that the winter males have delayed DA innervation.
12. Should axon length also be measured here (Fig 3)? It is not clear why the authors have switched to varicosity density. Also, a box should be drawn in the NAC cartoon to indicate the region that was sampled.
13. In Fig 3, Unc5c should be quantified to bolster the interesting finding that Unc5c expression dynamics are different between summer and winter hamsters. Unc5c mRNA experiments would also be important to see if similar changes are observed at the transcript level.
14. Fig 4. The peak in exploratory behavior in winter females is counterintuitive and needs to be better discussed. IN general, the light dark behavior seems quite variable.